

Week 7 Solve With Us



Augustin Louis Cauchy

(21 August 1789- 23 May 1857)

[[Augustin-Louis Cauchy](#) in 1821, followed by [Karl Weierstrass](#), formalized the **definition of the limit of a function**]

- Let $\{a_n\}$ be a sequence of real numbers, where $n \in \mathbb{N}$.
- $\{a_n\}$ is said to have a limit a , if the numbers a_n comes closer and closer to a , as n increases.
- If $\{a_n\}$ has a limit, then the sequence $\{a_n\}$ is said to be convergent sequence, otherwise it is said to be divergent.
- Some rules to find limits: Let $a_n \rightarrow a$ and $b_n \rightarrow b$, then
 - i) $a_n + b_n \rightarrow a + b$
 - ii) $ca_n \rightarrow ca$
 - iii) $a_n - b_n \rightarrow a - b$
 - iv) $a_nb_n \rightarrow ab$
 - v) $f(a_n) \rightarrow f(a)$, where f is any polynomial function.
 - vi) $\frac{a_n}{b_n} \rightarrow \frac{a}{b}$, if $b \neq 0$
 - vii) $c^{a_n} \rightarrow c^a$, for any real number c , where c^{a_n} is real number for each $n \in \mathbb{N}$.
 - viii) $\log_c(a_n) \rightarrow \log_c(a)$, if $a_n > 0$ for all $n \in \mathbb{N}$, and $a, c > 0$.
 - ix) $c_n \rightarrow a$, if $a_n \leq c_n \leq b_n$, for all $n \in \mathbb{N}$.

Question 1

Which of the following option(s) is(are) true?

[Hint: If the given function is of the form $\frac{f(x)}{g(x)}$ and $g(x) \rightarrow 0$, as $x \rightarrow a$. In this case, if possible, try to cancel the factor $(x - a)$ from both $f(x)$ and $g(x)$, and then try to evaluate the function at $x = a$.]

$$(1) \lim_{x \rightarrow -1} \frac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$$

$$(2) \lim_{x \rightarrow 0} \frac{x^2 - 6x - 7}{x^2 + 3x + 2} = -8$$

$$(3) \lim_{x \rightarrow 2} (x^3 + 4x^2 - 6x - 7) = 5$$

$$(4) \lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} = 1$$

- **For Option 1:** $\lim_{x \rightarrow -1} \frac{x^2 - 6x - 7}{x^2 + 3x + 2} = \lim_{x \rightarrow -1} \frac{(x + 1)(x - 7)}{(x + 1)(x + 2)} = \lim_{x \rightarrow -1} \frac{(x - 7)}{(x + 2)}.$
- **For Option 2:** To find the value of: $\lim_{x \rightarrow 0} \frac{x^2 - 6x - 7}{x^2 + 3x + 2},$ evaluate the given function at $x = 0,$ as the function is defined at $x = 0.$
- **For Option 4:** $\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x - 3)^2}{x - 3} = \lim_{x \rightarrow 3} (x - 3)$

The graph of some function is given below in the Figure M2W1G1. Choose the set of correct options w.r.t the given graph.

Question 2

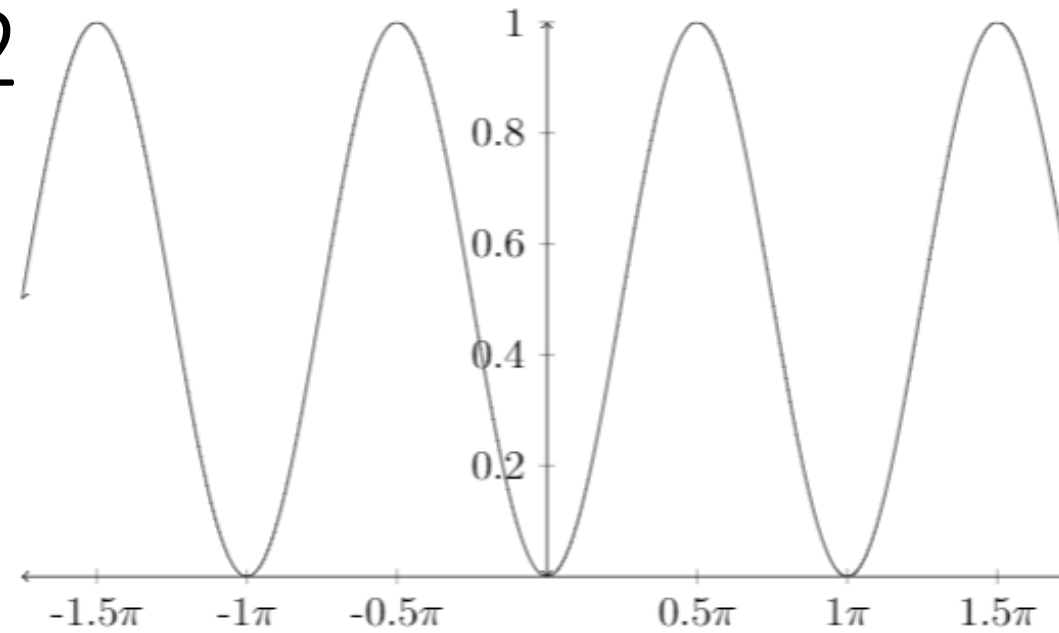


Figure: M2W1G1

- ☐ Option 1: Limit of the function as x tends to 0 is 1.
- ☐ Option 2: Limit of the function as x tends to 0 is 0.
- ☐ Option 3: Limit of the function as x tends to 0 is undefined.
- ☐ Option 4: There is a (unique) tangent at the point $x = \pi$, but not at $x = -\pi$.
- ☐ Option 5: There is a (unique) tangent at $x = \pi$, as well as at $x = -\pi$.
- ☐ Option 6: The given function is monotonically increasing in the interval $[-0.5\pi, 0.5\pi]$.
- ☐ Option 7: The given function is monotonically decreasing in the interval $[-0.5\pi, 0.5\pi]$.

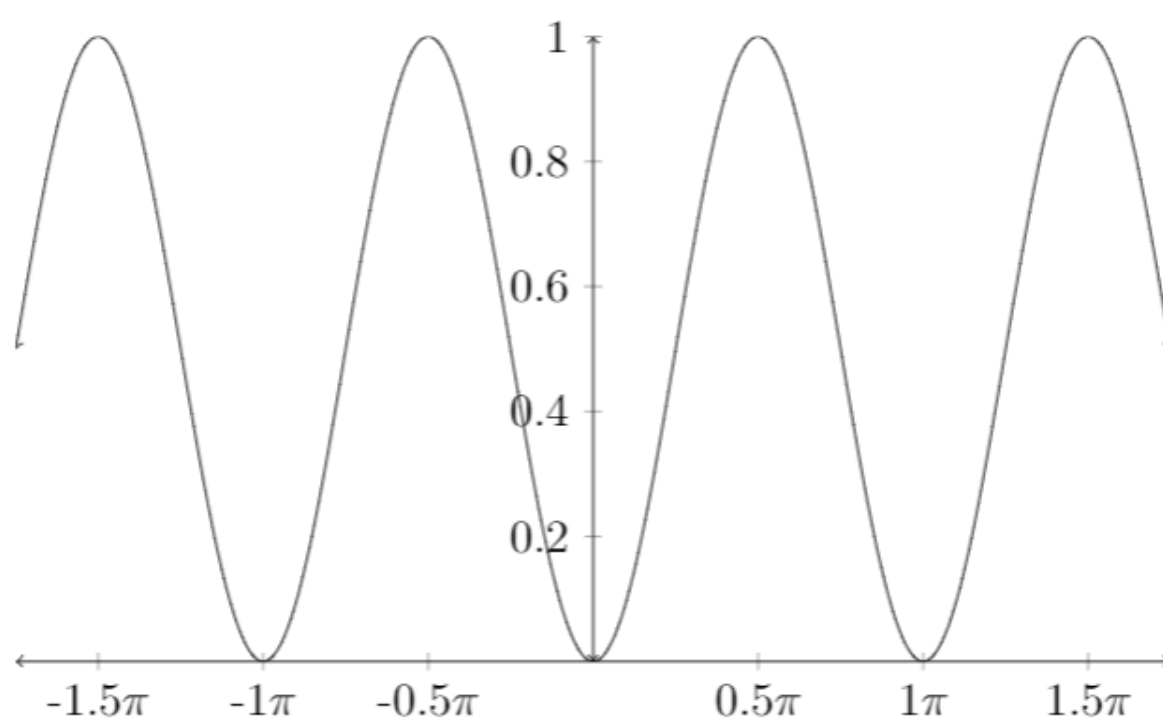


Figure: M2W1G1

- ☐ Option 1: Limit of the function as x tends to 0 is 1.
- ☐ **Option 2: Limit of the function as x tends to 0 is 0.**
- ☐ Option 3: Limit of the function as x tends to 0 is undefined.
- ☐ Option 4: There is a (unique) tangent at the point $x = \pi$, but not at $x = -\pi$.
- ☐ **Option 5: There is a (unique) tangent at $x = \pi$, as well as at $x = -\pi$.**
- ☐ Option 6: The given function is monotonically increasing in the interval $[-0.5\pi, 0.5\pi]$.
- ☐ Option 7: The given function is monotonically decreasing in the interval $[-0.5\pi, 0.5\pi]$.

Question 3

Some more things to remember..

- $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$

Using the given information, choose the set of correct options.

- ☐ Option 1: $2 \lim_{x \rightarrow 0} \frac{\log(1+x)}{\sin(x)} = 2.$
- ☐ Option 2: $2 \lim_{x \rightarrow 0} \frac{\log(1+x)}{\sin(x)}$ is undefined.
- ☐ Option 3: $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = 1.$
- ☐ Option 4: $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = 3.$
- ☐ Option 5: $\lim_{x \rightarrow 0} \frac{\sin(5x)}{x} = \frac{1}{5}$
- ☐ Option 6: $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = 1.$
- ☐ Option 7: $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = 5.$
- ☐ Option 8: $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = \frac{1}{25}.$
- ☐ Option 9: $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = \frac{1}{5}.$

Using the given information, choose the set of correct options.

- ☐ **Option 1:** $2 \lim_{x \rightarrow 0} \frac{\log(1+x)}{\sin(x)} = 2.$
- ☐ **Option 2:** $2 \lim_{x \rightarrow 0} \frac{\log(1+x)}{\sin(x)}$ is undefined.
- ☐ **Option 3:** $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x} = 1.$
- ☐ **Option 4:** $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x} = 7.$
- ☐ **Option 5:** $\lim_{x \rightarrow 0} \frac{\sin(7x)}{x} = \frac{1}{7}$
- ☐ **Option 6:** $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = 1.$
- ☐ **Option 7:** $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = 5.$
- ☐ **Option 8:** $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = \frac{1}{25}.$
- ☐ **Option 9:** $\lim_{x \rightarrow 0} \frac{e^{\frac{x}{5}} - 1}{\sin 5x} = \frac{1}{5}.$

Limits of some standard functions are given below:

- $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow \infty} x \left(\frac{\sqrt{2\pi x}}{x!} \right)^{\frac{1}{x}} = e$
- $\lim_{x \rightarrow \infty} \frac{x}{\sqrt[x]{x!}} = e$

Question 4: Find the value of $\lim_{n \rightarrow \infty} e^{\sqrt[n]{n!}} \left[\log \left(1 + \frac{6}{n} \right) - \frac{e^{\frac{1}{n}} - 1}{(\sqrt{2n\pi})^{\frac{1}{n}}} \right]$

$$\lim_{n \rightarrow \infty} e^{\sqrt[n]{n!}} \left[\log \left(1 + \frac{6}{n} \right) - \frac{e^{\frac{1}{n}} - 1}{\left(\sqrt{2n\pi} \right)^{\frac{1}{n}}} \right] = \lim_{n \rightarrow \infty} e^{\sqrt[n]{n!}} \left[\frac{\log \left(1 + \frac{6}{n} \right)}{\frac{6}{n}} \times \frac{6}{n} - \frac{\frac{e^{\frac{1}{n}} - 1}{\frac{1}{n}} \times \frac{1}{n}}{\frac{n \left(\sqrt{2n\pi} \right)^{\frac{1}{n}}}{\left(n! \right)^{\frac{1}{n}}}} \right]$$

$$= \lim_{n \rightarrow \infty} e \left[\frac{\log \left(1 + \frac{6}{n} \right)}{\frac{6}{n}} \times \frac{6}{\frac{n}{\sqrt[n]{n!}}} - \frac{\frac{e^{\frac{1}{n}} - 1}{\frac{1}{n}}}{n \left(\sqrt{2n\pi} / n! \right)^{\frac{1}{n}}} \times \frac{\frac{n \sqrt[n]{n!}}{n}}{\frac{n \sqrt[n]{n!}}{n}} \right]$$

Let $\frac{1}{n} = x$. As $n \rightarrow \infty, x \rightarrow 0$. Then the function becomes,

$$= e \left[\lim_{x \rightarrow 0} \frac{\log(1 + 6x)}{(6x)} \times \frac{6}{\lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}}} - \frac{\lim_{x \rightarrow 0} \frac{e^x - 1}{x}}{\lim_{n \rightarrow \infty} n \left(\frac{\sqrt{2n\pi}}{n!} \right)^{\frac{1}{n}}} \times 1 \right] = e \left[1 \times \frac{6}{e} - 1 \times \frac{1}{e} \right] = 6 - 1 = 5$$

- $\lim_{x \rightarrow 0} \frac{\log(1 + x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$
- $\lim_{x \rightarrow \infty} x \left(\frac{\sqrt{2\pi x}}{x!} \right)^{\frac{1}{x}} = e$
- $\lim_{x \rightarrow \infty} \frac{x}{\sqrt[x]{x!}} = e$

Question 5

Suppose a company runs three algorithms to estimate its future growth. Suppose the error in the estimation depends on the available number (n) of data as follows:

- Error in estimation by Algorithm 1: $a_n = \frac{n^2 + 5n}{6n^2 + 1}$.
- Error in estimation by Algorithm 2: $b_n = \frac{1}{8} + (-1)^n \frac{1}{n}$
- Error in estimation by Algorithm 3: $c_n = \frac{e^n + 4}{7e^n}$

where $n \in \mathbb{N} \setminus \{0\}$. Suppose the company has a large amount of data in hand (we can assume n tends to ∞). Using the information given above, answer questions

Which of the following statements is (are) correct?

- ☐ Error in estimation by Algorithm 1 will give the maximum error.
- ☐ Error in estimation by Algorithm 3 will give the minimum error.
- ☐ Both Algorithm 1 and Algorithm 2 will give the same error and that will be the maximum.
- ☐ Error in estimation by Algorithm 1 will be 0.166 approximately.
- ☐ Error in estimation by Algorithm 2 will be 0.125 approximately.

- Solution:

Error estimation from algorithm 1:

$$\text{Applying limit, } \lim_{n \rightarrow \infty} \frac{n^2 + 5n}{6n^2 + 1} = \lim_{n \rightarrow \infty} \frac{1 + \frac{5}{n}}{6 + \frac{1}{n}} = \frac{1}{6} = 0.166 \text{ (as } n \rightarrow \infty, \frac{1}{n} \rightarrow 0 \text{)}.$$

Error estimation from algorithm 2:

$$\lim_{n \rightarrow \infty} \frac{1}{8} + \frac{(-1)^n}{n} = \frac{1}{8} + \lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = \frac{1}{8} + 0 = 0.125$$

Error estimation from algorithm 3:

$$\lim_{n \rightarrow \infty} \frac{e^n + 4}{7e^n} = \lim_{n \rightarrow \infty} \frac{1 + \frac{4}{e^n}}{7} = \frac{1}{7} \approx 0.142$$

- Error in estimation by Algorithm 1 will give the maximum error.
- Error in estimation by Algorithm 3 will give the minimum error.
- Both Algorithm 1 and Algorithm 2 will give the same error and that will be the maximum.
- Error in estimation by Algorithm 1 will be 0.166 approximately.
- Error in estimation by Algorithm 2 will be 0.125 approximately.